

Problem

For the two-port in Fig. 19.94,

$$[h] = \begin{bmatrix} 16\ \Omega & 3 \\ -2 & 0.01\ \text{S} \end{bmatrix}$$

Find:

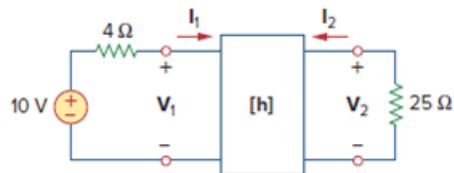
(a) V_2/V_1

(b) I_2/I_1

(c) I_1/V_1

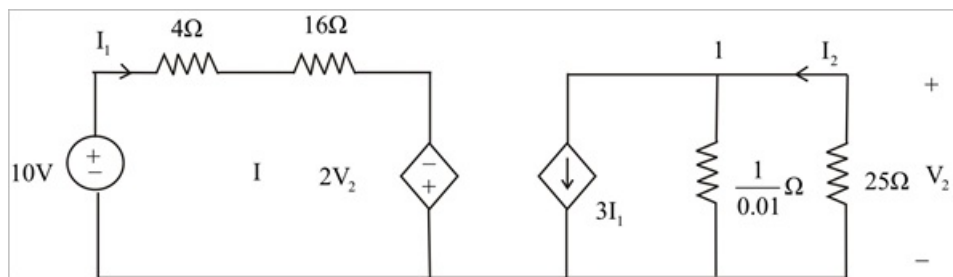
(d) V_2/I_1

Figure 19.94



Step-by-step solution

Step 1 of 3



Step 2 of 3

Applying current division at 1

$$I_2 = -3I_1 \times \left(\frac{1}{0.01} \right) \times \frac{1}{\frac{1}{0.01} + 25}$$

$$\frac{I_2}{I_1} = -2.4 = -2.4$$

$$V_2 = -3I_1 \times \left(\frac{1}{0.01} \parallel 25 \right) \Rightarrow \frac{V_2}{I_1} = \frac{-3(2500)}{125} = -60\ \Omega$$

Step 3 of 3

$$\text{In loop I} \quad 10 = 20I_1 - 2V_2$$

$$V_1 = 20I_1 - 2 \times 60I_1$$

$$\frac{V_1}{I_1} = 20 - 120 = -100$$

$$\frac{I_1}{V_1} = 0.01S$$

$$\frac{V_2}{V_1} = \frac{60I_1}{100I_1} = 0.6$$

$$a. \quad \frac{V_2}{V_1} = 0.6 \quad b. \quad \frac{I_2}{I_1} = -2.4 \quad c. \quad \frac{I_1}{V_1} = 0.01S \quad d. \quad \frac{V_2}{I_1} = 60\Omega$$